An Interactive Streaming Harness for Human Evaluation of Real-Time 3D Renderers

Connor Ding <czsding@stanford.edu>
Ishita Gupta <ishitagupta@stanford.edu>

1 Background and Setup

Novel View Synthesis (NVS) is an active area of research in 3D computer graphics. The goal is to reconstruct a 3D representation of a scene from a sparse collection of photographs taken from multiple camera angles. Optimization-based approaches that fit either differentiable neural fields (Mildenhall et al., 2020; Müller et al., 2022) or Gaussian primitives (Kerbl et al., 2023) have advanced the field substantially in both rendering quality and inference speed; we refer readers to Tewari et al. (2022) for a comprehensive survey.

For any 3D rendering method, defining "good quality" is open-ended: the final product is a visual artifact intended to be perceived by human eyes. The literature predominantly relies on image-based metrics — PSNR, MS-SSIM, and LPIPS (Zhang et al., 2018) — which measure the difference between ground-truth photographs held out from training and reconstructed images rendered from those held-out cameras. As numerical distances from ground truth these metrics are informative, but in the context of 3D rendering they have two important limitations:

1. The image-quality-assessment literature has argued for two decades that good PSNR or MS-SSIM scores do not necessarily translate to a good perceptual experience (Wang and Bovik, 2009). LPIPS partially addresses this by replacing pixel-level comparisons with learned deep features, but it remains a proxy for direct viewing and is comparatively expensive to compute.

2. Image-based metrics evaluate *static* images at a small, fixed set of held-out camera poses, whereas practitioners and end users actually *interact* with 3D scenes — moving the camera, zooming, and viewing continuously from arbitrary angles. Free navigation introduces failure modes (popping, floater drift, smoothness artifacts at oblique angles) that no static-image metric can detect.

To address this gap, we develop an interactive streaming harness for human evaluation of real-time 3D renderers. The system is a prototype, but it embodies a specific philosophy: the quality of a 3D rendering algorithm should ultimately be measured by how participants perceive it while interactively viewing it. A researcher provides one or two renderers together with a study configuration; the system streams those renderers to participants' browsers, collects per-aspect quality ratings (and, in two-method studies, an overall A/B preference together with aspect citations), and returns the raw per-participant data for the researcher to analyze.

The system's inputs are a researcher-provided rendering algorithm and its fitted 3D model, together with a study configuration (scenes, conditions, protocol). Its outputs are the data collected from each participant's session: (a) *subject ratings* produced through the rating UI — per-aspect 0–100 sliders, A/B preferences, and aspect citations; (b) *free-text feedback* entered in the post-session notes field; and (c) *mouse-movement trajectories* logged over the participant's viewing window. The

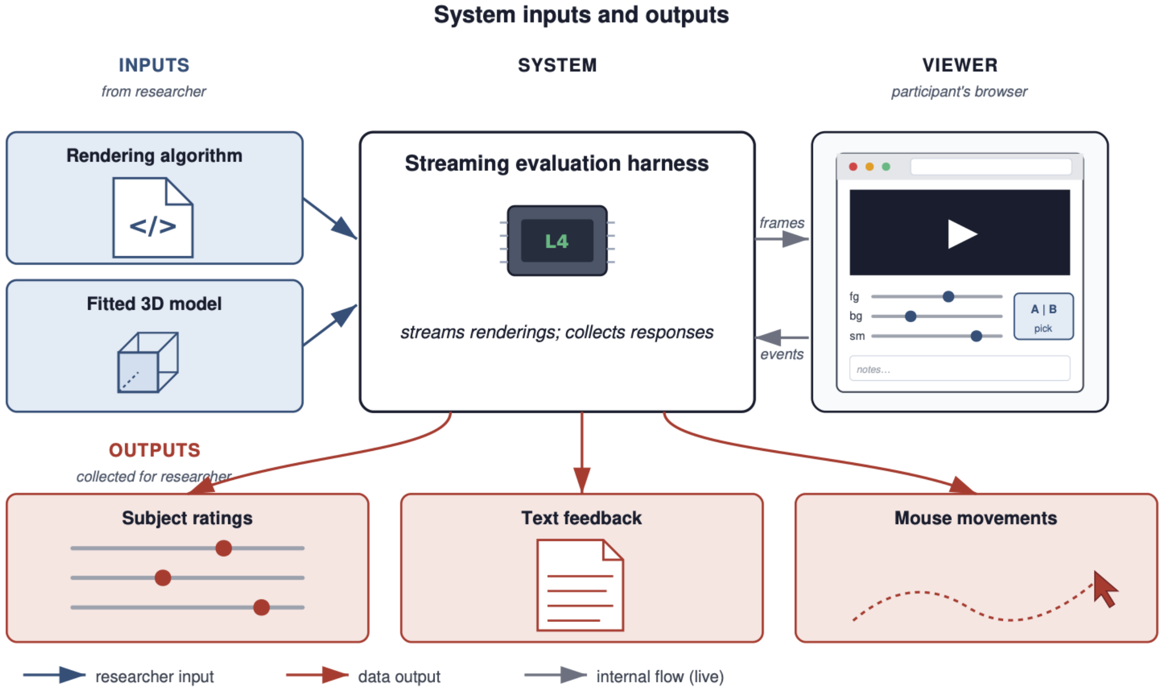

Figure 1: Conceptual view of system inputs and outputs. The researcher provides a rendering algorithm and a fitted 3D model; the system streams the resulting renderings to a participant's browser-based viewer and collects three categories of response data — subject ratings, free-text feedback, and mouse-movement trajectories — for the researcher to analyze.

first two are the immediate data products of the rating task; the third lets a researcher reconstruct *which views* of the scene each participant chose to inspect before issuing a rating. Figure 1 gives a high-level view of how the system sits between the researcher and the participant.

The three *main goals* are:

1. Measure a dimension of rendering quality that image-based metrics on held-out static frames cannot capture: the participant's *interactive perceptual experience* under continuous, free-roaming viewing. The collected data is intended as citable evidence a researcher can use to support claims about how their method actually behaves in interactive use, complementing rather than competing with the offline numbers already reported in the literature.

2. Offer a plug-and-play framework where a researcher can connect their renderer and trained model and run a human study with minimal infrastructure work.

3. Support both single-method subjective studies and two-method comparison studies, covering the two most common shapes of perceptual question a 3D-graphics researcher might ask.

The three explicit *non-goals* are:

1. Replace image-based numerical metrics — those metrics are cheap to compute, can be used as training losses, and remain the right tool for in-the-loop comparisons during method development.

2. Modify the underlying rendering algorithms — we provide a platform for evaluation, not for method improvement.

3. Provide statistical analysis or visualization of the collected data — the harness delivers raw per-participant response data; how that data is summarized, correlated against offline metrics, or visualized is the researcher's call.

Two practical *constraints* shape the system's design:

1. *Participants evaluate from their own machines*, which we cannot assume to have GPUs capable of running an NVS renderer. The renderer must therefore run on a remote GPU-equipped instance, with the rendered output delivered to the participant's browser over the network. We host the active renderer(s) and supporting services on an AWS L4 GPU instance.

2. *The framework should be agnostic to the underlying rendering algorithm.* Different NVS methods use different rendering stacks — CUDA-accelerated hash grids for i-NGP, a custom rasterizer for Nexels, the `gsplat` differentiable rasterizer for 3DGS — and we want to support not just these three but whatever NVS method a future researcher brings. A framework that asked each renderer to ship its native output format to the browser would have to keep pace with every method's evolving toolchain. Streaming a common video format (H.264) abstracts that diversity away: every renderer emits the same output regardless of its internal representation.

These constraints together motivate the streaming architecture described in Section 2.

To exercise the harness, we run two case studies of the kind a graphics researcher might reach for when validating their own model. The first is a single-viewpoint study — participants view one rendering at a time and rate it. The second is a two-viewpoint study — participants compare two NVS models side by side and select an overall preference along with the aspects that drove that choice. The focus is on the system rather than the findings of either case study; provided the resulting data are coherent and non-degenerate, the harness has done its job.

A good evaluation harness must satisfy two parties. On the researcher side, it should run on the GPUs and trained models a graphics researcher already has, and return interpretable, meaningful data without bespoke infrastructure work. On the participant side, each session should feel smooth and bug-free, with imperceptible streaming latency and a clear rating interface. We make both criteria precise in Section 3. The crux is twofold, and the methodological half is the larger contribution. *Methodologically*, we have to design a questionnaire and a study protocol that meaningfully capture interactive perceptual experience — a design problem with no off-the-shelf answer in the 3D-rendering literature, where every existing instrument either reduces to an image-metric proxy or asks the participant to rate a static frame. *Infrastructurally*, we have to deliver real-time renderings of multiple high-end NVS methods to a remote browser at participant-imperceptible latency, on a single L4 GPU, with a renderer interface generic enough that adding a new method does not require modifying the study logic. Section 2 describes both; Section 3 reports the two case studies.

2 Approach

We start from existing third-party renderer implementations for the three NVS methods we evaluate — `pyngp` for i-NGP (Müller et al., 2022), `diff-nexel-rasterization` for Nexels (Rong et al., 2025), and `gsplat` (Ye et al., 2024) for 3D Gaussian Splatting (Kerbl et al., 2023). What those projects do not ship is the instrument: there is no agreed-upon questionnaire for asking a participant how good a real-time 3D rendering feels, and no off-the-shelf platform that runs such a questionnaire while the participant interactively views a live render. The contribution of this work is, in order of importance, the design of that questionnaire (§2.1) and the streaming harness that lets it run (§2.2).

2.1 Questionnaire design

Decomposing "quality". A single overall "how good is this rendering" rating is hard to interpret across scenes and methods: it averages over different failure modes, and the participant's internal weighting is opaque. We instead decompose perceived quality into three sub-aspects that we elicit separately:

1. *Foreground detail* — sharpness and fidelity on the nominal subject of the captured photographs.
2. *Background detail* — sharpness, fidelity, and absence of floater artifacts on the surrounding regions, which NVS methods tend to struggle with more than the foreground.
3. *Motion smoothness* — continuity of the rendering during free navigation. This is the one aspect that *requires* interactive viewing to evaluate at all.

The first two are familiar from still-image quality work; the third is the explicit reason this study exists as a streaming harness rather than an image grader. The three axes also align with the kinds of design tradeoffs an NVS researcher actually reasons about — model capacity allocated to subject vs. surroundings, and the speed cost of additional parameters or splats.

Sliders rather than Likert. Each aspect is rated on a continuous 0–100 slider with the endpoints labeled ("poor" and "excellent") and the midpoint unlabeled. We chose continuous sliders over a 5-point Likert scale for two reasons: (i) sliders avoid the anchor-driven clustering that compresses Likert distributions toward one or two modes, and (ii) on a 0–100 scale a single within-participant slider can be correlated directly against an offline metric on the same axis, which a 5-bucket categorical cannot.

Two-method comparison: independent viewports, forced A/B, aspect citation. For the two-method comparison the participant sees two *independently orbitable* viewports side by side, with method identities hidden behind anonymous badges (only the `test` initials hook reveals them, and is never used in real sessions). Independence is deliberate: a single baked camera trajectory would dictate which artifacts the participant ever sees, but pilot sessions confirmed that participants want to *probe* the scene — zooming on suspicious regions, viewing the subject from oblique angles — and the artifacts they probe (popping, smoothness, floaters at grazing angles) are precisely the ones a fixed trajectory would either hide or showcase. After exploring a pair, the participant fills out per-side foreground / background / smoothness sliders, then makes a forced A/B overall preference ("A", "B", or explicit tie), then selects from a multi-select list *which* aspect(s) shaped that choice. The slider-then-A/B redundancy is intentional: we collect both a graded within-method rating and a categorical between-method judgment per trial, and the aspect multi-select converts an aggregate preference into something mechanistic — which aspect, on which scene, swung the choice.

Single-method convergence: ratings in isolation, reference anchor. The convergence case uses the same primitives in a different shape. The participant sees one viewport, rendering one stimulus (here: a 3DGS checkpoint at a given training-iteration count) at a time; the active stimulus is swapped without a process restart, so the only thing that changes between consecutive stimuli is the model state. We deliberately rate each stimulus in isolation rather than as A/B against the previous one, since the stimuli are not always commensurable on a single axis of merit and a forced choice would discard graded information. Per stimulus we collect two sliders — overall quality and motion smoothness — on the same 0–100 scale as the comparison case. To give the participant a ground-truth anchor for what the scene *should* look like, we play a low-frame-rate slideshow of the training photographs for the scene once before its rating block; an NVS rendering on its own provides no reference for "how detailed should this actually be".

2.2 Streaming platform

The streaming pipeline exists to let either questionnaire above run while the participant views a live render rather than a pre-baked trajectory, despite participants connecting from their own (typically GPU-less) laptops, and despite the underlying renderers having different native rendering stacks — the two constraints flagged in §1. The platform is, by intent, the smaller half of the contribution: it is what allowed us to ship the questionnaire, not a research artifact in itself.

Pipeline. Each renderer is wrapped in a thin process (`interactive_renderer.py`, `nexels_renderer.py`, `gsplat_renderer.py`) that consumes camera-pose deltas on a per-method WebSocket and pushes H.264 over RTSP to a `mediamtx` instance, which relays to the browser over WebRTC (WHEP). A single browser-facing WebSocket multiplexer (`ws_mux.py`) routes pose-delta messages to the right renderer based on URL path, and a static HTTP server (`serve.py`) hosts the viewer pages and accepts rating submissions. All server-side processes colocate on a single AWS L4 GPU instance. Figure 2 shows the resulting data flow.

Plug-and-play seam. The principal design choice on the platform side is the contract every renderer must satisfy: (a) accept camera-pose deltas as JSON messages on a per-method WebSocket port; (b) push H.264 over RTSP at the configured target resolution and frame rate (1440×810@30 for the comparison case, 1920×1080@60 for the convergence case); and (c) optionally accept a `swap_state` message used by the convergence case to flip the active checkpoint without restarting the renderer process. Any process that meets this contract drops in by registering a route in `ws_mux.py` and a corresponding RTSP path in `mediamtx.yml`. The browser never has to know which underlying method is active — the same viewer HTML works for any renderer that honors the contract, and the marginal cost of adding a third method is a few lines of configuration rather than a viewer rewrite.

What did not work. Two earlier designs are preserved in `archive/`. We initially considered a fully pre-rendered evaluation built on canned camera trajectories rendered to MP4 (`archive/start_noninteractiv` but pilot sessions surfaced exactly the central limitation that motivated the streaming approach: participants could not probe the artifacts they wanted to talk about. We then tried a single-renderer interactive pipeline built on Xvfb plus each renderer's native UI, with `event_relay.py` translating browser WebSocket events into `xdotool` keystrokes (`archive/start_interactive.sh`). This worked end-to-end but end-to-end latency was high (the screen-capture path added tens of milliseconds), and the design did not generalize to running two renderers concurrently for a side-by-side comparison because each one demanded its own X display. The current architecture —

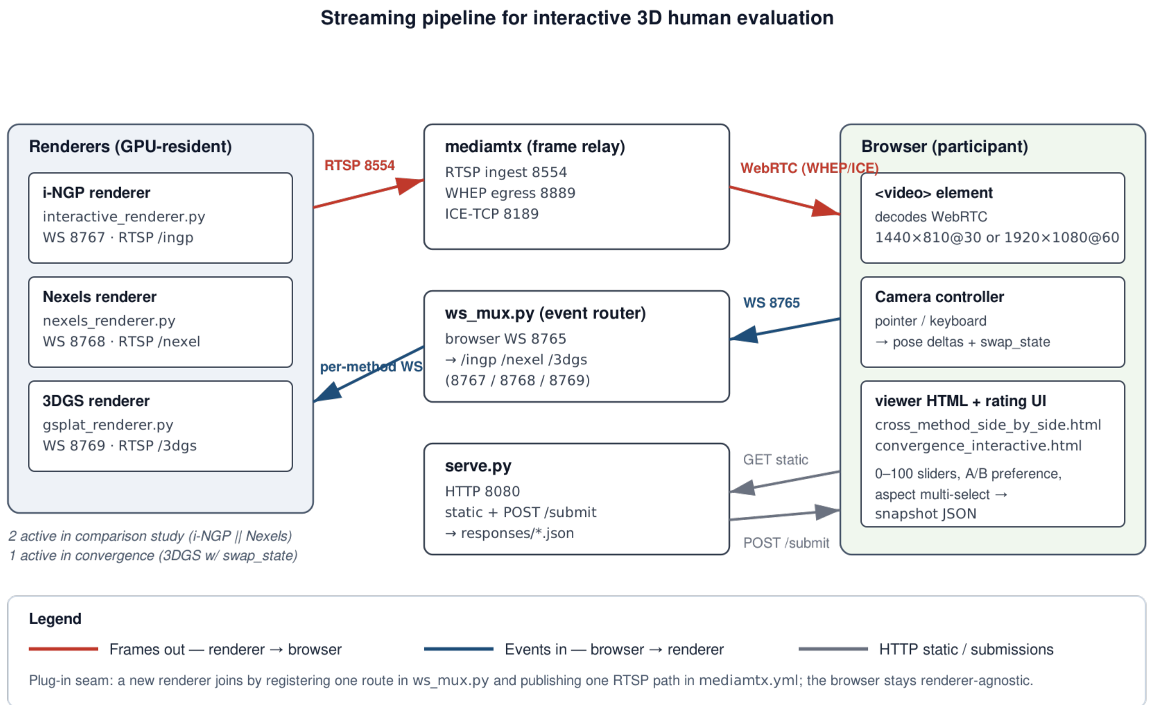

Figure 2: Streaming pipeline. The three renderers conform to a common contract (RTSP push for frames out, per-method WebSocket for events in); `ws_mux.py` and `mediamtx` between them keep the browser renderer-agnostic.

renderers that push H.264 directly to mediamtx, browser receives via WebRTC, event mux that routes per-method WebSockets — replaced both of these and is substantially simpler.

3 Evaluation and Results

3.1 Definition of success

A useful evaluation harness has to satisfy two parties, as flagged at the end of §1. We restate the two criteria here in concrete, checkable form.

Researcher-side success. The harness should give a graphics researcher data they can act on without engineering the data pipeline themselves. Concretely: (a) every participant who starts a session leaves a snapshot JSON on disk, including sessions that ended early; (b) the resulting data has informative structure — within-participant slider spread is non-trivial, A/B preferences are not all ties, and aspect citations are distributed across the three options rather than degenerate on one; and (c) the data shape is general enough that a researcher can apply analyses of their own choosing (correlations against offline metrics, within-trial regressions, inter-rater agreement) without bespoke parsing.

Participant-side success. The session should feel smooth and easy to complete. Concretely: (a) interactive latency is below the threshold at which participants notice it as “laggy”; (b) both renderers sustain their target frame rate over the full session without sustained dips; (c) no participant abandons mid-session for technical reasons; and (d) the rating UI is clear enough that no experimenter intervention is required.

Participant-side success is harder to measure than researcher-side success because it is partly experiential: systems telemetry can confirm that no frames were dropped, but it cannot confirm that the participant felt the session was pleasant. We rely on telemetry as an indirect proxy for now, and plan to administer a brief post-session self-report survey (in preparation) covering perceived smoothness, clarity of the rating UI, and fatigue (§3.5). The dual-party framing also clarifies what kind of failure invalidates what kind of conclusion: a participant-side failure (frame stalls, UI confusion) propagates into researcher-side measurements and invalidates them, while a researcher-side data-shape "failure" — sliders all pegged at one end, for instance — could itself be a valid measurement of a real ceiling effect rather than a harness bug.

3.2 Case studies

We exercised the harness with two case studies. Both ran on a single AWS GPU instance with an NVIDIA L4 (24 GB VRAM, `g6.xlarge`), hosting all server-side processes (`mediamtx`, `ws_mux`, `serve.py`, and the active renderers) on the same host. Participants connected from their own machines over the public internet; we did not constrain network conditions or display characteristics, and we discuss the resulting variance in §3.6. Across both studies we recruited 19 unique participants from the authors' friends and labmates; eleven of the twelve case-B participants had previously completed case A. A case-A session lasted a median of 9 minutes (mean 10); a case-B session a median of 5 minutes. We collected $n = 18$ complete sessions in case study A and $n = 12$ in case study B; recruitment is closed for this submission.

Case study A: comparison (i-NGP vs Nexels). Two NVS methods — i-NGP and Nexels — rendered side-by-side at a matched storage budget (≈14 MB per scene; see Table 1) over six mip-NeRF-360 scenes (`bicycle`, `counter`, `garden`, `kitchen`, `room`, `stump`) plus one practice scene (`bonsai`). The two methods stream concurrently to two side-by-side viewports at 1440×810@30. Method identity is hidden on the viewport badges (only the `test` initials hook reveals it, and is never used in real sessions). The seven scene pairs are presented in a per-participant random order; side assignment of i-NGP and Nexels is coin-flipped per pair; the first pair (`bonsai`) is the practice trial and is excluded from analysis. The questionnaire shape — per-side sliders, forced A/B, aspect multi-select — is described in §2.1.

Case study B: convergence (3DGS at four checkpoint counts). A single method (3D Gaussian Splatting) trained for varying numbers of iterations (1k, 3k, 7k, 30k) on two Tanks-and-Temples scenes (`train`, `truck`) — eight stimuli total. All eight checkpoints are GPU-resident (≈1.4 GB combined, 210k–3.8M splats per checkpoint), so swapping the active checkpoint is a constant-time pointer update. The renderer streams 1920×1080@60. Before each scene's rating block, the participant watches a low-frame-rate slideshow of the training photographs as a ground-truth reference. Scene order is randomized; iteration order within a scene is fixed ascending so the reference frame for *better* stays stable across the four levels.

Offline-metric baselines. For each scene–method pair in case study A and each scene–checkpoint pair in case study B we computed PSNR, SSIM, and LPIPS on the held-out test split for that scene (mip-NeRF-360 hold-every-eighth-view convention for case study A; gsplat validation split for case

Table 1: Case study A: *offline metrics* at the `small` size setting (≈14 MB per scene). PSNR / SSIM / LPIPS are computed on the held-out mip-NeRF-360 test split with a consistent LPIPS-AlexNet backbone; FPS is from each method's bench script over 200 cycled test-camera frames at 1297×840. Bold marks the per-scene winner on each column. `bonsai` is the practice scene and is excluded from the mean.

Scene	Method	PSNR↑	SSIM↑	LPIPS↓	FPS	MB
bicycle	i-NGP	21.93	0.462	0.592	**62.0**	14.5
	Nexels	**22.21**	**0.539**	**0.424**	29.5	**14.4**
counter	i-NGP	25.31	0.757	0.272	**61.5**	**13.5**
	Nexels	**26.42**	**0.833**	**0.216**	30.0	14.4
garden	i-NGP	23.13	0.514	0.456	**74.9**	**12.0**
	Nexels	**23.46**	**0.636**	**0.322**	34.0	14.4
kitchen	i-NGP	27.12	0.767	0.214	**96.2**	**12.6**
	Nexels	**27.68**	**0.861**	**0.167**	31.9	14.4
room	i-NGP	28.47	0.844	0.225	**69.8**	15.0
	Nexels	**29.15**	**0.877**	**0.207**	29.8	**14.4**
stump	i-NGP	21.68	0.473	0.574	27.0	16.0
	Nexels	**23.88**	**0.604**	**0.398**	**32.9**	**14.4**
mean	i-NGP	24.61	0.636	0.389	**65.2**	13.9
	Nexels	**25.47**	**0.725**	**0.289**	31.4	14.4

Table 2: Case study A: *per-scene human ratings* (n = 18 participants). **fg**, **bg**, **sm** are mean 0–100 slider scores for foreground, background, and rendering smoothness. **Votes** is the number of participants who preferred that method on that scene; tie counts are bicycle 3, counter 1, garden 4, kitchen 2, room 1, stump 1. Bold marks the per-scene winner on each column.

Scene	Method	fg↑	bg↑	sm↑	Votes
bicycle	i-NGP	73.3	46.4	66.6	2
	Nexels	**80.6**	**68.3**	**72.5**	**13**
counter	i-NGP	**68.6**	54.9	**55.4**	6
	Nexels	67.9	**59.2**	48.1	**10**
garden	i-NGP	71.3	62.3	**56.9**	7
	Nexels	**76.8**	**74.3**	45.9	7
kitchen	i-NGP	**77.6**	**62.3**	**66.4**	**14**
	Nexels	58.4	43.1	38.6	2
room	i-NGP	**58.6**	**59.2**	**57.5**	**10**
	Nexels	52.8	54.6	55.8	6
stump	i-NGP	60.7	52.9	**63.9**	5
	Nexels	**76.5**	**53.7**	59.4	**11**
mean	i-NGP	68.4	56.3	**61.1**	44
	Nexels	**68.9**	**58.9**	53.4	**49**

study B). LPIPS uses the AlexNet backbone in both cases for consistency. Tables 1 and 3 list these per-stimulus offline numbers alongside the corresponding mean slider scores.

3.3 Systems results

The harness met its interactivity targets in every session collected to date. Both renderers in case study A sustained the target 30 fps at 1440×810 throughout; the 3DGS renderer in case study B sustained 60 fps at 1920×1080 across all four checkpoint levels; no session was abandoned due to a renderer stall, dropped WebRTC stream, or submission failure; and checkpoint swaps in case study B introduced no measurable visual hitch. The per-method FPS columns of Tables 1 and 3 give the single-renderer ceilings at the offline benchmark resolution (1297×840 / gsplat eval). Per-scene FPS detail and the end-to-end latency budget are deferred to Appendix A.

3.4 Data informativeness

This subsection addresses the researcher-side success criterion: does the harness produce data with the shape and signal required for downstream analysis? We deliberately keep the discussion descriptive rather than inferential. The substantive claims that one could make *from* this data — about i-NGP vs Nexels, about the perceptual return on additional 3DGS training, about how PSNR tracks human preference — are properly the researcher's to make. Our role here is to verify that the harness has not produced degenerate data.

Three observations support the informativeness claim:

First, the slider distributions are spread. Table 2 shows mean foreground/background/ smoothness scores in case study A ranging from 38 to 81 across scene–method pairs, with no floor or ceiling

Table 3: Case study B. Offline metrics and per-stimulus mean human ratings ($n = 12$) for the convergence-study stimuli (3DGS at four checkpoint counts on Tanks-and-Temples `train` and `truck`). LPIPS is AlexNet (gsplat default), internally consistent within this table. FPS is `ellipse_time` at the gsplat eval resolution and is therefore an upper bound on the interactive frame rate at 1920×1080.

Scene	Iters	PSNR↑	SSIM↑	LPIPS↓	FPS	# Splats	fg↑	bg↑	sm↑
train	1k	17.06	0.584	0.550	433.7	0.21M	48.8	30.9	72.5
	3k	18.75	0.650	0.413	254.1	0.49M	63.7	41.3	71.2
	7k	19.73	0.721	0.298	130.8	1.05M	78.3	51.1	80.2
	30k	21.44	0.812	0.146	77.4	1.81M	84.2	64.2	79.2
truck	1k	20.94	0.725	0.326	437.5	0.21M	43.2	48.0	66.8
	3k	22.65	0.800	0.207	144.0	1.14M	58.4	66.2	82.8
	7k	23.87	0.845	0.143	67.6	2.51M	77.5	72.0	82.9
	30k	25.04	0.871	0.095	45.5	3.79M	81.3	71.8	60.1

effect. Convergence-case slider means (Table 3) similarly span 31 to 84 across the eight stimuli. Within-participant variance, while not shown in the tables, is non-trivial: participants regularly assigned different scores to different aspects of the same condition.

Second, A/B preferences in case study A are heterogeneous across scenes (Figure 3). The aggregate 44/49/12 (i-NGP / Nexels / tie) is close to even, but the per-scene distribution is anything but: `bicycle` swings 13–3–2 toward Nexels (sign test $p = 0.007$); `kitchen` swings 14–2–2 toward i-NGP ($p = 0.004$); the remaining four scenes sit closer to the aggregate average. The data discriminates strongly between scenes — exactly the kind of structure that lets a downstream analysis say something interesting.

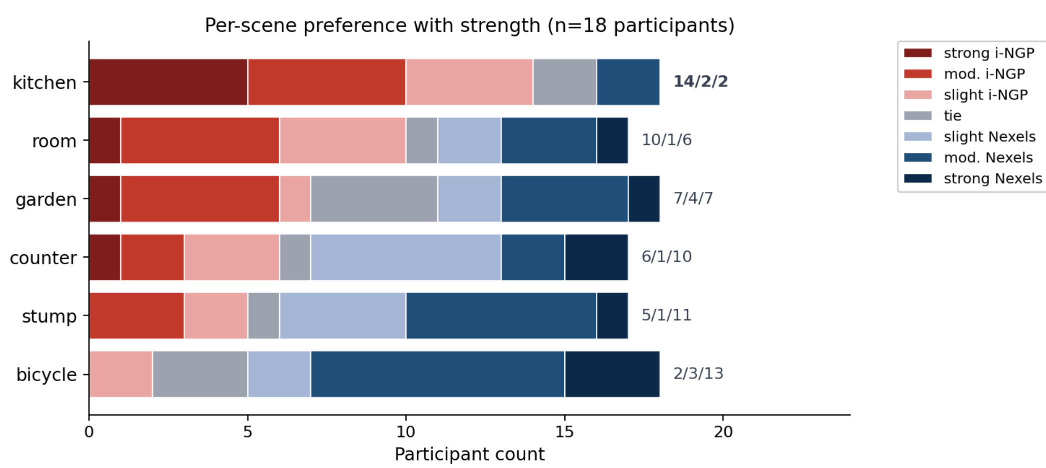

Figure 3: Per-scene A/B preference distribution from case study A ($n = 18$). Each bar segments the 7-point preference scale into strong / moderate / slight i-NGP (red shades), tie (gray), and slight / moderate / strong Nexels (blue shades), so the bar's colour spread encodes preference *strength* in addition to direction. Scenes ordered by net Nexels share, ascending. The cross-scene heterogeneity is the indication that the harness produces analyzable data; we do not interpret the per-scene rankings here.

Third, convergence-case slider trajectories across iteration counts show non-trivial patterns

(Figure 4). On the smaller scene (`train`), foreground and background ratings rise monotonically across all four checkpoints (foreground 49 → 84; background 31 → 64), with smoothness flat after 7k. On `truck`, the picture is more interesting: foreground saturates between 7k and 30k (77 → 81), background flattens (72 → 72), and *smoothness drops 23 points* from 82.9 at 7k to 60.1 at 30k — a non-monotone trajectory that the data captures cleanly even at $n = 12$. The four checkpoint levels are clearly distinguishable, and the data admits straightforward visual comparison against the offline LPIPS curve overlaid on the figure.

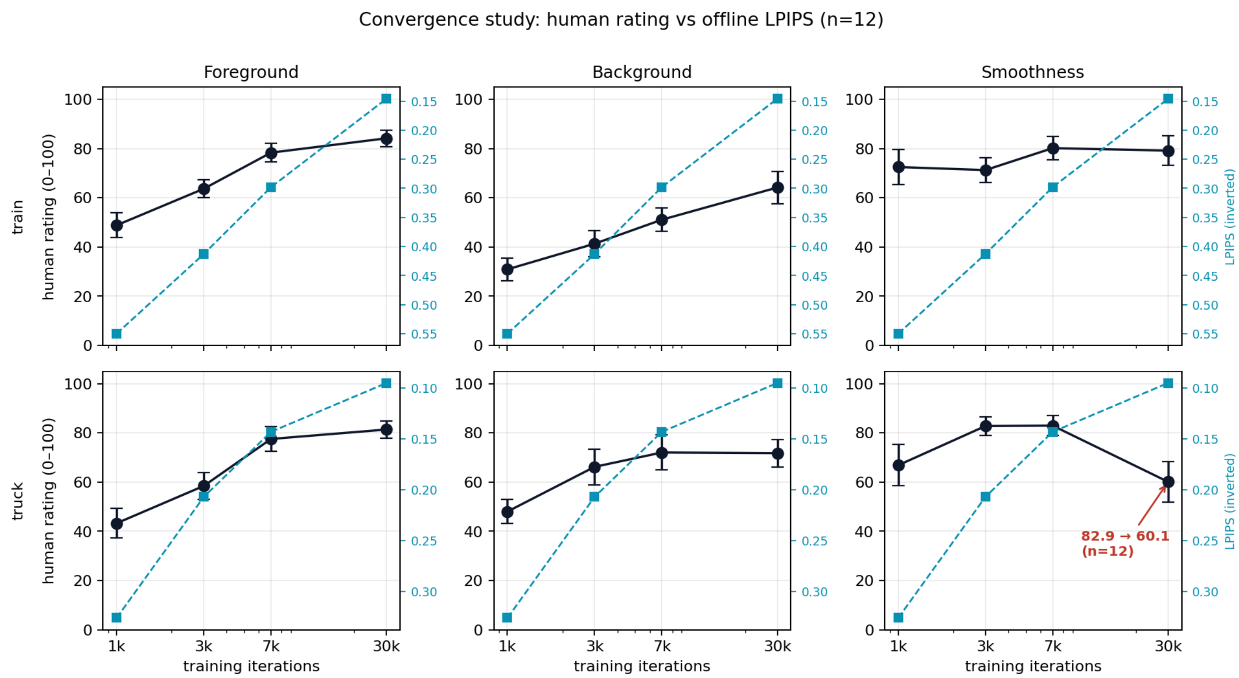

Figure 4: Per-aspect mean human rating ($n = 12$ participants, error bars are SEM) and offline LPIPS (teal dashed, axis inverted) across the four 3DGS checkpoints. Top row: `train`. Bottom row: `truck`. Columns are the three slider aspects.

These are descriptions of the data, not claims about NVS methods. A researcher analyzing this data is the appropriate party to make substantive claims; our role is to verify that the data has the structure needed to support them. Appendix B illustrates the kinds of analyses the data admits — Spearman correlations between A/B preference and offline-metric gap, Mann–Whitney U tests for aspect-citation effects, and (once $n = 18$ is large enough) Kendall's W or ICC$(2, k)$ for inter-rater agreement.

3.5 Participant feedback

This subsection addresses the participant-side success criterion via two channels: indirect indicators from session logs, and a brief post-session self-report survey administered to each participant.

Indirect indicators (from session logs). Every participant who loaded the viewer completed the session through to submission: there were no mid-session abandonments in either case study. Median session duration was approximately 9 minutes in case study A and 5 minutes in case study B — consistent with our piloting estimate. No experimenter intervention was required during any session, and no participant reported a stall, freeze, or unresponsive UI in the post-submission notes field.

Post-session self-report ($n = 9$). After each session we administer a five-question Likert survey (1–5 scale) covering overall experience, mental fatigue, perceived session length, instruction clarity, and ease of use of the interactive viewer; the questionnaire itself is documented in Appendix C. Figure 5 shows the response distribution for each question. The satisfaction survey was introduced partway through recruitment so its sample ($n = 9$) lags behind the rating data; numbers will be updated as additional survey responses come in.

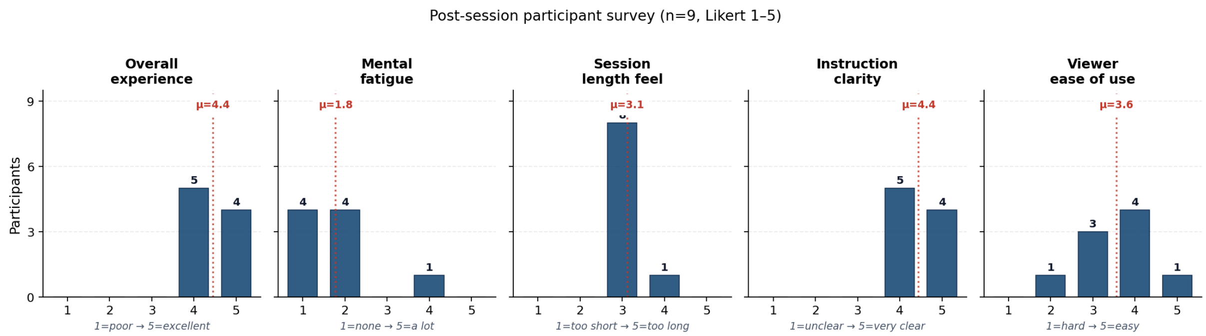

Figure 5: Distribution of post-session participant survey responses ($n = 9$), one histogram per question. Bar heights are participant counts; the dotted red line on each panel marks the mean. Anchor conventions: overall (1 = poor, 5 = excellent); fatigue (1 = none, 5 = a lot); length (1 = too short, 3 = just right, 5 = too long); clarity (1 = unclear, 5 = very clear); ease (1 = hard, 5 = easy).

The signal is positive across the board. Overall experience and instruction clarity both sit at mean 4.4 on the 5-point scale, with all responses ≥ 4 on both. Eight of nine participants rated session length as "just right" (3); the ninth rated it 4 ("a bit too long"). Mental fatigue is low (1–2) for eight of nine respondents, with one outlier at 4. The weakest dimension is ease of use of the interactive viewer (mean 3.56) — the 2 came from the same participant whose row also shows elevated fatigue (4), suggesting that for at least one participant the viewer UI was itself a source of cognitive load rather than just the rating task. With $n = 9$ we report these patterns descriptively rather than test them.

Free-response feedback. After each session participants could leave optional free-text comments. Reading across them, participants reported overall positive experience — *"really impressed by the different spaces it could capture"* · *"VERY CLEAR INSTRUCTIONS!"* · *"really liked this one — clear how much better things were improving over time"* — with common suggestions for improvement including a reset-view button, less-sensitive camera zoom, and consistent camera-drag direction. These align with the lower viewer ease-of-use rating in the survey and are concrete UI affordances we would prioritise in a follow-on iteration of the system.

3.6 Limitations

The most important limitation remains sample size. At $n = 18$ and $n = 12$ the statistical tests above are reasonable for the per-scene and per-stimulus signals we surface, but underpowered for finer-grained between-aspect comparisons; we report effect sizes alongside p-values where possible. Beyond sample size, several threats to validity warrant explicit acknowledgment.

Display variance. Participants connected over the public internet from uncalibrated monitors. Differences in display brightness, gamut, and refresh rate introduce noise we cannot correct for. The case study A design — both methods rendered to the same participant in the same session — partially controls for this, but case study B (stimuli presented sequentially) does not.

Network variance. WebRTC adapts bitrate dynamically based on network conditions, which means a participant on a slower network may be evaluating a different visual stimulus than one on a faster connection. We do not currently log per-session bitrate; for a future version this should be instrumented and either gated or stratified.

Selection bias. Participants were recruited from the authors' academic environment and self-selected for interest in 3D rendering; generalization to the broader population is not claimed.

Order and learning effects. Case study B fixes iteration order as ascending within scene, trading order-effect contamination for a stable reference frame for "better". Case study A randomizes scene order and A/B side assignment per pair. A first-pair learning effect may still inflate ratings on early scenes; the `bonsai` practice trial is intended to absorb most of this but is not a complete control.

Method-blinding integrity. The viewer hides method identities behind anonymous badges. We did not collect post-hoc data on whether participants could guess which side was which, which would be the standard check on blinding in a perception study.

Participant-side measurement is indirect. As described in §3.5, our current participant-side success indicators are entirely indirect. The planned post-session survey is the appropriate fix, but it is not yet in the field for the current data.

4 Team Responsibilities

This was a two-person project. The system, studies, and writeup were ultimately joint products; the table below records who took primary responsibility for each piece.

Responsibility	Lead
System design and implementation (assisted by Claude Code)	
Overall architecture and streaming pipeline	Connor
Renderer wrappers (i-NGP, Nexels, 3DGS)	Connor
WebSocket multiplexer and submission server	Ishita
Viewer HTML and questionnaire UI	Ishita
3DGS checkpoint training for case study B	Connor
mediamtx / RTSP / WebRTC plumbing	Connor
Studies	
Questionnaire and protocol design	Joint
Participant recruitment	Joint
Session administration	Joint
Post-session satisfaction survey design	Ishita
Analysis and writing	
Offline-metric collection and tables	Connor
Statistical analyses and figures	Connor
Report drafting (§1–3)	Joint
Final-presentation slides	Ishita
Feedback and editorial passes on report and slides	Ishita

The studies, the recruitment, the discussion of findings, and the editorial passes on the writeup were genuinely shared, and the table's joint rows reflect that — we both signed off on every artifact the project ships.

References

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics (SIGGRAPH)*, 42(4), 2023.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision (ECCV)*, 2020.

Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. Instant neural graphics primitives with a multiresolution hash encoding. *ACM Transactions on Graphics (SIGGRAPH)*, 41(4), 2022.

Victor Rong, Jan Held, Victor Chu, Daniel Rebain, Marc Van Droogenbroeck, Kiriakos N Kutulakos, Andrea Tagliasacchi, and David B Lindell. Nexels: Neurally-textured surfels for real-time novel view synthesis with sparse geometries. *arXiv preprint arXiv:2512.13796*, 2025.

Ayush Tewari, Justus Thies, Ben Mildenhall, Pratul Srinivasan, Edgar Tretschk, Wang Yifan, Christoph Lassner, Vincent Sitzmann, Ricardo Martin-Brualla, Stephen Lombardi, Tomas Simon, Christian Theobalt, Matthias Nießner, Jonathan T. Barron, Gordon Wetzstein, Michael Zollhöfer, and Vladislav Golyanik. Advances in neural rendering. *Computer Graphics Forum (Eurographics State of the Art Report)*, 41(2), 2022.

Zhou Wang and Alan C. Bovik. Mean squared error: Love it or leave it? A new look at signal fidelity measures. *IEEE Signal Processing Magazine*, 26(1):98–117, 2009.

Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Letian Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for Gaussian splatting. *arXiv preprint arXiv:2409.06765*, 2024.

Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

A Systems details

Per-method FPS detail. The single-renderer FPS columns of Tables 1 and 3 give the per-method bench-rate ceilings at the offline benchmark resolution (1297×840). i-NGP is on average 2–3× faster than Nexels at the `small` model size (mean 65.2 vs. 31.4 fps), with the exception of `stump` where i-NGP's frame rate collapses to 27 fps for reasons related to that scene's geometry (large depth range, many empty rays). For 3DGS, FPS scales inversely with splat count: 434 fps at the 1k checkpoint (210k splats) drops to 77 fps at the 30k checkpoint (1.8M splats) on `train`, and 437 → 45 fps on `truck` (3.8M splats). For case study B these numbers are all well above the 60 fps interactive target at 1920×1080, so frame-rate variability inside a session was not a confound; for case study A, both renderers cleared the 30 fps target with substantial headroom on five of six scenes.

Pending telemetry. [To be added: end-to-end input-to-photon latency breakdown (render, encode, network, decode, display); GPU memory residency over the course of a case-study-A session; per-renderer frame-time distribution under cohabitation versus single-renderer operation.]

B Example analyses on case-study A data

The analyses below are illustrative of the kind of work a researcher might do on data the harness collects. We present them not as scientific claims about i-NGP or Nexels but as evidence that the data shape supports such analyses. All numbers are computed on the $n = 18$ case-study-A dataset.

Preference vs. offline-metric gaps (Spearman correlations). We computed Spearman correlations between A/B preference (sign convention: + Nexels preferred, − i-NGP preferred) and the per-scene gap on each offline metric (sign-aligned to "Nexels has the better offline number"). Table 4 reports the full set; Figure 6 plots the scene-level FPS–preference relationship.

Table 4: Spearman correlation between A/B preference (+ Nexels / − i-NGP) and per-scene offline-metric gap. Trial level pools all (participant, scene) rows; scene-mean level averages within scene first.

Offline-metric gap	Trial-level ρ ($n = 105$)	p	Scene-mean ρ ($n = 6$)	p
PSNR	−0.06	0.56	−0.03	0.96
SSIM	+0.02	0.85	+0.20	0.70
LPIPS (AlexNet)	**+0.34**	**< 0.001**	**+0.83**	**0.04**
FPS	**+0.35**	**< 0.001**	+0.77	0.07

Aspect-citation effects (Mann–Whitney U). After each A/B choice, participants ticked which of the three aspects (foreground, background, smoothness) influenced that choice. Table 5 reports Mann–Whitney U tests on whether citing a given aspect changes the strength and direction of preference. The decomposition is interpretable: background citations correlate with stronger preferences toward Nexels; smoothness citations with preferences toward i-NGP; foreground citations are roughly neutral on direction. We do not over-interpret these effects; they exemplify the within-trial decomposition the aspect multi-select makes possible.

Table 5: Aspect-citation effects on A/B preference. Each row tests whether citing that aspect changes preference strength (left) or preference direction (right). Mann–Whitney U test, two-sided.

Aspect	|pref| cited (n)	not cited (n)	p	signed cited	not cited	p
Foreground	1.88 (57)	1.31 (48)	**0.003**	+0.30	−0.15	0.18
Background	1.95 (59)	1.20 (46)	**< 0.001**	**+0.63**	**−0.59**	**< 0.001**
Smoothness	2.09 (34)	1.39 (71)	**< 0.001**	**−0.62**	**+0.44**	**0.009**

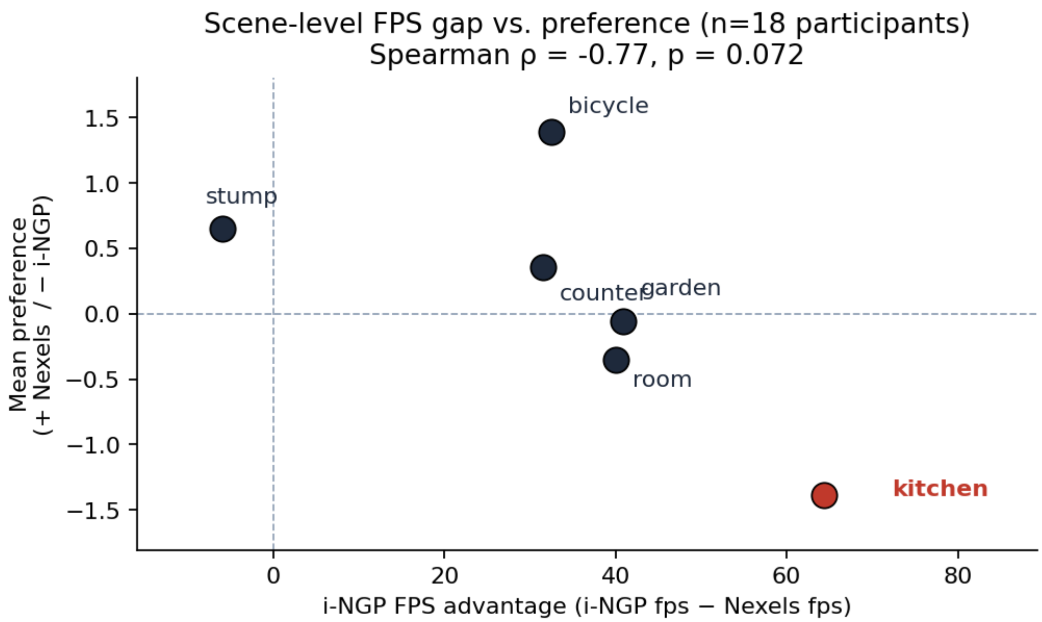

Figure 6: Per-scene mean preference (+ Nexels / − i-NGP) plotted against i-NGP's FPS advantage on that scene ($n = 18$ participants). `kitchen` is highlighted. Scene-level $\rho = -0.77$ ($p = 0.07$); trial-level $\rho = -0.35$ ($p < 0.001$, $n = 105$). The sign convention places i-NGP preference in the lower half-plane.

Per-scene breakdown. Table 6 resolves Table 5 one level further by splitting each aspect's citation count by both scene and the direction the citer chose. The column-total row exposes an asymmetry that the aggregate table cannot: Nexels-choosing citers invoke smoothness only 12 times across all six scenes, against 33–38 for foreground and background; i-NGP-choosing citers invoke all three aspects nearly equally (24/21/22). Participants who chose Nexels were almost always doing so on the strength of *what they saw*; participants who chose i-NGP did so on the strength of what they saw *and* how it moved.

Instrument internal consistency. Per-row slider gaps (Nexels rating minus i-NGP rating on each aspect) correlate with the same row's A/B preference at $\rho = 0.65$ (foreground), 0.71 (background), and 0.60 (smoothness), all $p < 10^{-10}$ at $n = 105$. When participants rate the methods differently on a continuous slider, that flows through cleanly to their forced choice on the same trial — the two instruments measure compatible things.

Inter-rater agreement. At $n = 18$ standard inter-rater agreement metrics (Kendall's W on A/B preference per scene; ICC$(2, k)$ on per-aspect slider scores) become computable with usefully narrow confidence intervals. We defer the specific numbers to a follow-up pass.

C Post-session participant satisfaction questionnaire

We administer the following five-question Likert survey (Google Forms, 1–5 scale) after each rating session. The survey is intentionally short — five multiple-choice items — to keep response rate high;

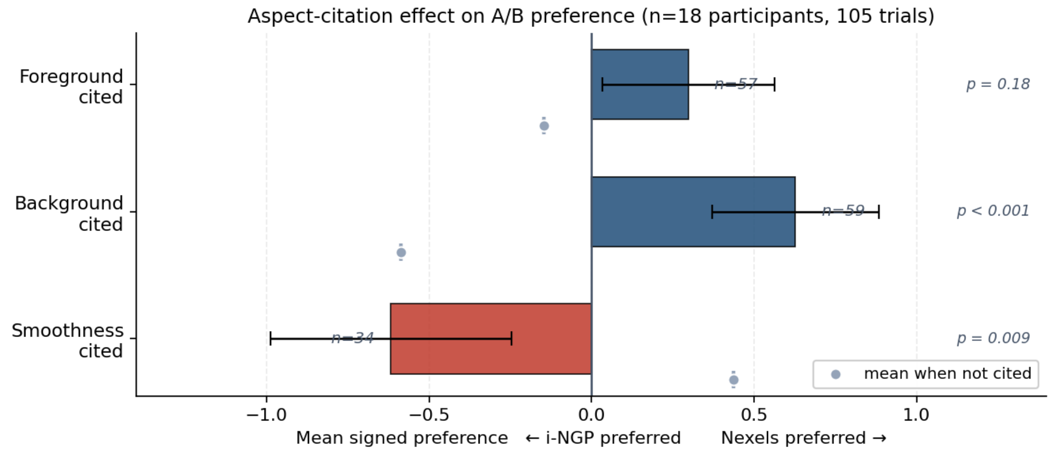

Figure 7: Visualization of the direction column of Table 5: mean signed A/B preference on trials where each aspect was cited (bars; red = pulls toward i-NGP, blue = pulls toward Nexels), with the corresponding not-cited mean shown as a gray marker for contrast. Citing background pulls preference toward Nexels; citing smoothness pulls it toward i-NGP; foreground citations are roughly neutral on direction. Error bars are SEM.

qualitative comments are already captured by the rating viewer's notes field, so this post-session instrument is purely quantitative.

1. *Overall, how was your participant experience?* (1 = poor → 5 = excellent)
2. *By the end of the session, how mentally fatigued did you feel?* (1 = not at all → 5 = a lot)
3. *How long did the length of the session feel?* (1 = too short, 3 = just right, 5 = too long)
4. *How clear were the instructions and questions for the tasks?* (1 = unclear → 5 = very clear)
5. *How easy was it to use the interactive viewer?* (1 = hard → 5 = easy)

The five items together cover the four facets of participant-side success defined in §3.1: overall experience (Q1); cognitive load and pacing (Q2, Q3); task interpretability (Q4); and UI quality (Q5). Aggregated results to date appear in §3.5.

Table 6: Per-scene aspect citations split by A/B preference direction. Each cell counts trials where the row's aspect was cited *and* the participant chose the column's method. No tied participant cited any aspect, so a third column group is unnecessary. Bold marks the largest cell per scene within the winning column group.

	i-NGP citers			Nexels citers		
Scene	fg	bg	sm	fg	bg	sm
bicycle	1	0	2	8	**11**	3
counter	3	4	3	**7**	6	1
garden	3	2	4	2	**7**	3
kitchen	**8**	5	7	2	1	0
room	**8**	5	5	4	5	2
stump	1	5	1	**10**	8	3
total	24	21	22	33	**38**	12